EDUVERSE

Developed By:

Shreyansh Dhanani - U02001625

Pratham Modi - U01994470

Bhargav Kakadiya- U01996543

Jay Kakadiya – U01987717

Guided By:

Kshitij Sharma

Institution:

Pace University, Seidenberg School of Computer Science & Information Systems

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1. TEAM MEMBER ROLES AND RESPONSIBILITIES

Team Member	Role	Responsibilities
Shreyansh Dhanani	Full Stack Developer / Tech Lead	- Backend architecture design (NestJS, ExpressJS) - Database schema design and management - Code review and technical documentation
Pratham Modi	Backend Developer	- API development and integration - System deployment and performance optimization - Code review and technical documentation
Bhargav Kakadiya	Frontend Developer	- Frontend development (ReactJS) - User experience testing - Frontend documentation
Jay Kakadiya	UI/UX Designer	- UI/UX design and implementation - Component architecture

2. PROBLEM STATEMENT

- Problem Definition: Existing online learning platforms often fail to address the diverse needs of students, course providers, and institutions in a cohesive manner. Institutions lack comprehensive tools for managing subscriptions and providing students with access to high-quality educational resources. Course providers struggle with limited content management capabilities, and super administrators are unable to oversee the platform effectively due to fragmented management tools. Eduverse seeks to resolve these issues by offering an integrated platform that supports all user roles effectively. By streamlining course management, institutional collaboration, and administrative oversight, Eduverse ensures an efficient, accessible, and engaging learning experience for all stakeholders.

3. PROJECT DESCRIPTION

- The goal of the comprehensive online learning platform Eduverse is to provide a smooth educational ecosystem that links universities, super administrators, course suppliers, and students.
- Motivated by websites such as Coursera and Udemy, Eduverse facilitates communication between students and academic institutions by allowing:
- Universities could buy subscription plans and give their students free access to courses.
- Course providers to produce, oversee, and profit from instructional materials
- Students can obtain certifications and take advantage of a variety of learning opportunities.
- Super administrators to supervise user management, analytics, and platform operations
- By providing an integrated solution with role-based access, institutional communication capabilities, and centralized administrative control, the platform seeks to address the disjointed experience of current solutions.

4. PERSONAS

1. The Student – Maya Chen

Role: Final-year Computer Science Student / Aspiring Full-Stack Developer
Demographics: 21 years old, tech enthusiast, active learner

Goals:

- Gain industry-relevant skills (e.g., React, Vue.js) beyond university syllabus
- Build a portfolio with projects and verifiable certificates
- Learn flexibly with on-demand video and progress tracking

Frustrations:

- University curriculum is too theoretical and outdated
- Hard to find structured, high-quality courses from trusted experts
- Needs credible proof of learning for employers

How the Platform Helps:

- Browse expert-led courses with clear descriptions and difficulty levels
- Access video lessons, quizzes, and assignments in one place
- Track enrolled courses and progress via dashboard
- Earn verifiable certificates upon completion

2. The University Administrator – Dr. Anya Sharma

Role: University Administrator
Demographics: 45 years old, PhD in Education, tech-savvy, budget-conscious

Goals:

- Quickly expand course offerings with industry-relevant content
- Ensure external courses meet academic quality standards
- Centrally monitor student enrollment, progress, and certification
- Provide value through cost-effective subscription plans

Frustrations:

- Manual, slow approval process for external courses
- No reliable way to verify student certificates
- Lack of unified overview of student engagement across courses

How the Platform Helps:

- Discover and approve vetted courses from qualified providers
- Monitor registered students, enrollment, and completion status
- Verify certificate authenticity through secure issuance system
- Access analytics and manage university-level access and reporting

3. The Course Provider – Alex Rivera

Role: Freelance Developer / Tech Educator

Goals:

- Monetize expertise by creating structured, paid courses
- Reach a wider audience, including university students
- Easily upload, organize, and manage course content

Frustrations:

- Building and maintaining a standalone course site is complex
- Difficulty attracting students and competing with big platforms
- Needs automated handling of payments, enrollments, and access

How the Platform Helps:

- Use simple tools to upload videos, PDFs, quizzes, and set pricing
- Manage courses and monitor student progress via provider dashboard
- Rely on integrated payment and subscription system for transactions
- Gain exposure through university partnerships and admin approvals

5. TECHNOLOGIES

React.JS (Frontend)

- Chosen for its component-based architecture, virtual DOM for performance, and extensive ecosystem. Enables building a dynamic, responsive user interface with reusable components across different user roles.

Next.Js, Nest.JS (Backend)

- Selected for its modular architecture, TypeScript support, and dependency injection. Provides excellent structure for scalable server-side applications, ideal for Eduverse's multi-role system.

Express.JS + Node.JS

- Lightweight foundation that works seamlessly with NestJS. Offers non-blocking I/O for handling concurrent user sessions – critical for high-traffic learning platforms.

PostgreSQL

- Chosen for ACID compliance, robust relational capabilities, and complex query support. Essential for managing relationships between users, courses, enrollments, subscriptions, and transactions.

GitHub

- Enables version control, branch management, and collaborative development. Provides clear history of contributions and facilitates code reviews.

6. Algorithms

1. Role-Based Access Control (RBAC) Algorithm

- **Purpose:** Secure user authentication & permission management
- **Logic:** Hierarchical role verification (Super-Admin → University-Admin → Course-Provider → Student)

2. Collaborative Filtering Algorithm

- **Purpose:** Personalized course recommendations
- **Logic:** User-based filtering (students with similar profiles) + Content-based filtering (course attributes)

3. Domain Matching Algorithm

- **Purpose:** University verification & student association
- **Logic:** Email domain extraction + pattern matching against registered university domains

4. Payment Gateway Integration Algorithm

- **Purpose:** Secure transaction processing & subscription management
- **Logic:** Tokenization + encryption + 3D secure authentication

5. Weighted Scoring Algorithm

- **Purpose:** Student progress tracking & completion status
- **Logic:** Multi-factor weighted calculation (Lessons 40% + Quizzes 30% + Assignments 20% + Time 10%)

6. Approval Workflow Algorithm

- **Purpose:** Course submission & review process
- **Logic:** State machine (Pending → Review → Approved/Rejected) with notifications

EDUVERSE - AI FEATURE INTEGRATION

1. AI-Driven Dynamic Assessment System:

Project Overview: This module serves as the core evaluation engine of our online learning platform, leveraging the power of Generative AI to replace static question banks. By integrating the OpenAI (ChatGPT) API, the system generates unique, high-quality quizzes in real-time based on the user's selected technology stack, such as JavaScript, Python, or Cloud Computing. Beyond just testing, the AI performs a deep analysis of user responses to provide personalized feedback, transforming a simple test into a comprehensive learning experience.

Core Functionalities:
Intelligent Topic Selection: Users can define their "Area of Exam" from a wide range of technical domains. The system then contextualizes the difficulty level based on the user's profile.

On-the-Fly Quiz Generation: Instead of pulling from a database, the system sends a structured request to ChatGPT to formulate original Multiple Choice Questions (MCQs) with plausible distractors.

AI-Powered Feedback Loop: Once the quiz is submitted, the AI doesn't just score the user; it analyzes the logic behind incorrect answers and suggests specific areas for improvement.

Seamless JSON Integration: The backend sanitizes AI responses into structured data formats, ensuring a smooth and bug-free UI rendering for the learner.

2. Intelligent AI Learning Assistant (Chatbot):

Project Overview: The AI Chatbot is an integrated real-time assistant designed to support learners throughout their educational journey on the platform. Powered by the ChatGPT API, this bot acts as a 24/7 personal tutor that helps users clarify doubts, understand complex quiz results, and prepare for their "Area of Exam" (e.g., JavaScript) through interactive dialogue. It bridges the gap between passive testing and active learning by providing instant, conversational explanations for any technical concept.

Core Functionalities:

Pre-Exam Guidance: Users can ask the chatbot for a summary of key topics or "Cheat Sheets" before attempting an AI-generated quiz.

Post-Quiz Analysis: The chatbot can ingest the user's specific quiz mistakes and explain the underlying logic in a simplified, conversational manner.

Context-Aware Support: The bot is "aware" of the user's selected subject, allowing it to provide code snippets, debugging tips, and best practices relevant to the exam area.

Interactive Doubt Solving: Beyond the quiz, users can prompt the bot with specific questions like "Explain Closures in JavaScript with an example," receiving instant, high-quality responses.

7. PROJECT SCHEDULE – CADENCE

Task ID	Task Description	Assigned To	Status	Deliverable
T-001	Project ideation and concept finalization	All Team Member	✅ Complete	Project brief document
T-002	Market research and competitor analysis	Jay Kakadiya	✅ Complete	Comparative analysis report
T-003	Technology stack selection	Shreyansh Dhanani	✅ Complete	Stack decision document
T-004	GitHub repository setup	Pratham Modi	✅ Complete	Repository: Eduverse
T-005	Initial project documentation	Bhargav Kakadiya	✅ Complete	README, project overview
T-006	Sprint 0 planning and task allocation	Pratham Modi	✅ Complete	Sprint 0 backlog
T-007	Presentation slide deck preparation	Shreyansh Dhanani	✅ Complete	Sprint 0 slides
T-008	Presentation video recording	All Team Member	✅ Complete	7 – 10minute video

8. TEAM WORKING AGREEMENT

Meeting Schedule:

- **Daily Stand-up:** 15 minutes via Discord/Teams (10:00 AM)
- **Sprint Planning:** Every Monday, 30 minutes
- **Sprint Review/Retrospective:** Every Friday, 45 minutes
- **Ad-hoc Meetings:** Scheduled as needed via mutual agreement

Task Commitment & Deadlines:

- All tasks assigned during sprint planning must be updated on the project board within 24 hours
- Tasks should be updated with status (To Do, In Progress, Blocked, Done) daily
- If a deadline cannot be met, communicate at least 24 hours in advance

Work Distribution

- Work is distributed based on role responsibilities and individual capacity
- Both members commit to equal contribution and shared ownership
- Code reviews are mandatory before merging to main branch
- Documentation is part of "done" for every task

Communication Protocols

- **Primary Channel:** WhatsApp group (urgent matters)
- **Secondary Channel:** GitHub Issues (task-specific discussions)
- **Response Time:** Within 4 hours during working hours (10:00 AM – 7:00 PM)
- **File Sharing:** Google Drive for documents, GitHub for code
- All major decisions must be documented in writing

9. RETROSPECTIVE

What Went Well:

- **Open Communication:** The team members had open communication, which facilitated the exchange of ideas, opinions, and feedback. The team encouraged collaboration, which ensured that everyone was on the same page and working towards the same objectives.

- **Well-Defined Roles and Responsibilities:** The team had well-defined roles and responsibilities, which ensured that everyone knew what was expected of them. This clarity helped the team to work efficiently and effectively, as everyone knew what they needed to do to achieve the projects objectives.

- **Agile Methodology:** The team's adoption of an Agile methodology was a strength, as it allowed the team to be responsive to changes and adapt to new requirements quickly. The approach enabled the team to work in sprints and focus on prioritized tasks, which resulted in an efficient and effective development process.

What Needs Improvement:

- **Better Time Management:** The team could improve their time management skills by setting more realistic goals and deadlines. The team could also prioritize tasks and set achievable milestones to ensure that they deliver quality work within the given timeframe.

- **Enhanced Documentation:** The team could improve their documentation practices by ensuring that project-related information is accurately documented and easily accessible. This includes project plans, meeting minutes, progress reports, and technical documentation.

- **Continuous Learning:** The team could focus on continuous learning and skill-building to keep up with the latest technology and development trends. The team could also consider incorporating training and development opportunities into the project plan to help team members learn new skills and technologies.

Goal For Next Sprint:

- Enhance our collaboration by sharing knowledge and resources, seeking feedback, and supporting each other. We will also aim to create a positive and inclusive work environment to promote creativity and innovation.

Wiki Link :
https://github.com/shreyanshdhanani/eduverse/wiki

THANK YOU

Shreyansh Dhanani
Pratham Modi
Bhargav Kakadiya
Jay Kakadiya